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Supplementary Materials for

Cortical folding scales universally with surface area and thickness, not number of neurons

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Materials and Methods

We compiled data on numbers of neurons, cortical surface area, volume of the cortical gray matter, cortical thickness and folding index for one single cortical hemisphere for 74 species: 38 spread over 7 clades from our own previous studies (*14-21*) and 45 from others (including 6 species that overlapped with our dataset [*1,22-24*]). In our own dataset, all data (including numbers of cortical neurons) were available for 24 of the 38 species; the dataset from other studies lacks numbers of neurons, but had all other variables available for 38 species (6 of which overlap with our dataset). The entire dataset for which cortical surface area, cortical thickness and folding index were available consists of 55 species (Supplementary Table S1). Numbers of neurons in our dataset were obtained using the isotropic fractionator (*33*), which has been shown to yield estimates that are similar to those obtained with stereology (*34*).

To test our model, we made paper balls of sheets of A4 office paper of different surface areas and thickness by dividing sheets in half in a geometric series, and stacking different numbers of sheets before crumpling them. Once each crumpled paper ball relaxed, its diameter was measured along the three principal orthogonal axes and used to calculate the exposed surface area of the equivalent ellipsoid. The folding index for each paper ball was calculated as the ratio between the total surface area of the unfolded paper and the equivalent surface area of the crumpled paper ball. Allometric relationships (power laws) for biological cortices and crumpled paper balls were calculated using JMP 9.0 (SAS, USA).

Our model for the folding of self-avoiding membranes subjected to internal bulk forces is based on the axonal growth/contraction response to traction forces; that is, it is based on the plastic rather than elastic response to the longitudinal forces generated by cortical growth. Like Van Essen's longitudinal tension-driven folding hypothesis (*35*), it is also a connectivity-driven folding mechanism, although one based on differential elongation rather than longitudinal tension of axons.

A consequence of this hypothesis is that axonal tension in a (quasi-)static cortex is the (quasi-)equilibrium state in which the external applied force and the internal tension of each axon are balanced such that the axon's size remains constant. This corresponds to a minimum energy state, and implies that in a fully developed or growing cortex the direction of measured stress in a small volume of white matter should be along the average direction of its constituting axonal fibers (i.e., parallel to the principal component of the diffusion tensor in MRI), and its magnitude should be a function of their directional coherence. This is compatible to what has been found in literature (*36*).

In a microscopic scale, our model is based on the way axons respond plastically (i.e., changing their permanent lengths over an extended period [*26,27*]) to the external longitudinal forces that should arise from the expansion of the total cortical surface (from increases in neuronal numbers and size). Above a certain threshold, axons elongate at linearly with applied force; but they contract if the applied force falls below the threshold. Thus, if l is the length of an axon, its elongation rate is

$$\frac{dl}{dt} = v(\sum F_{ext} - F_0)$$

The contraction force F_0 can be considered the gradient of a linear potential. Assuming the same holds true for the external forces, one can write

$$\frac{dl}{dt} = -v\nabla(U_{ext} - \mathrm{F}_0 l)$$

Mathematically, this equation describes a tethered particle in a viscous Newtonian fluid, subject to a constant internal longitudinal force and external forces. Assuming the external forces change slowly, the particle will eventually settle in a position that minimizes its total potential energy, where the forces will be in equilibrium and therefore *l* will be constant. This equilibrium state correspond to the configuration in which axonal length *l* is minimized, subject to the external constraints from which the external forces are derived (an unconstrained minimization of length would of course imply *l*=0). Assuming further that F_0 is proportional to the cross-sectional area of the axon, so that the internal energy associated with each axon is proportional to its volume, then the cortex will assume a configuration that minimizes the total white matter volume, subject to external constraints. The proportionality constant will have dimension of a (negative) pressure. Since a growing cortex will also have to exert work against the cerebrospinal fluid, we can define a volumetric energy for the cortex that is simply

$$E_V = (p + p_{cs})V_w$$
$$E_V = (p + p_{cs})(V - TA_T)$$

where p_{cs} is the pressure of the cerebral-spinal fluid.

To this volume energy term one must add a surface term due to the energy associated with the shape of the cortical surface, so as to insure that it never intersects itself, plus whatever other surface energy terms ones cares to include. The self-avoiding term takes the form

$$E_S[\vec{s}, n] = \int d^2\vec{x}_1 d^2\vec{x}_2 \delta^2\big(\vec{s}(\vec{x}_1) - \vec{s}(\vec{x}_2)\big)\Lambda_T\left(n(\vec{x}_1) - n(\vec{x}_2)\right)$$

where $\vec{x}$ is the internal 2-dimensional coordinates of a point in the cortical surface and $\overrightarrow{\{s(\vec{x}), n(\vec{x})\}}$ is its 3-dimensional mapping into respectively the parallel and normal components of its position in space with respect to the cortical surface, and $\Lambda_T(u) = T - 2|u|$ if -T/2<u<T/2 and zero otherwise (*37*).

The second biologically-inspired element in the model is not strictly necessary to guarantee gyrification, but provides a mechanism whereby it occurs hierarchically, starting with the formation of the deepest sulci connecting more distant regions and progressing to increasingly smaller features. Diffusion magnetic resonance imaging analyses of axonal bundles in white matter have shown that they typically cross each other in orthogonal directions (*38*). In a spherical cortex (for simplicity of discussion) of radius R, an axon connecting points in the surface separated by an angle θ will process radially down distance r, then follow an arc parallel to the surface of length (R-r) θ, then up r again back to the surface. The total axonal length will thus be

$$l = 2r + (R - r)\theta$$

By checking the sign of its first derivative for different values of θ, it can be shown that l(r) will be minimized for r=0 if $\theta<116^o$, and for r=R if $\theta>116^o$. Thus, axons will naturally segregate between deep long-distance connections, and shorter-range, shallow connections. As the cortical surface increases, the resulting forces will make it expand preferentially in directions that will elongate the least amount of axons for the same gain in area. Thus, regions well connected by deep fibers will remain close, forming sulci, and those with fewer such connections will expand into lobular gyri; as the curvature inside the gyri increases, axons for which shallow connections were energetically preferred will be pulled by the resulting forces into now less energetic deep fiber configurations within the gyrus. Further cortical area expansion will in this manner form a new generation of sulci and gyri, as cortical expansion continues this process is iterated. All sulci and gyri will originally form at the same scale, but the surface expansion means that earlier-forming structures will become larger and deeper and will connect more distant regions, and that the larger structures will themselves be composed of smaller ones iteratively, down to the minimal, fundamental scale at which they are originally formed. This structure of nested self-similar structures precisely characterizes a fractal object. Of course, the fractal nature of the cortex is an approximation that only remains valid for scales greater than the fundamental scale as calculated below (otherwise, the cortex would have infinitely small gyri).

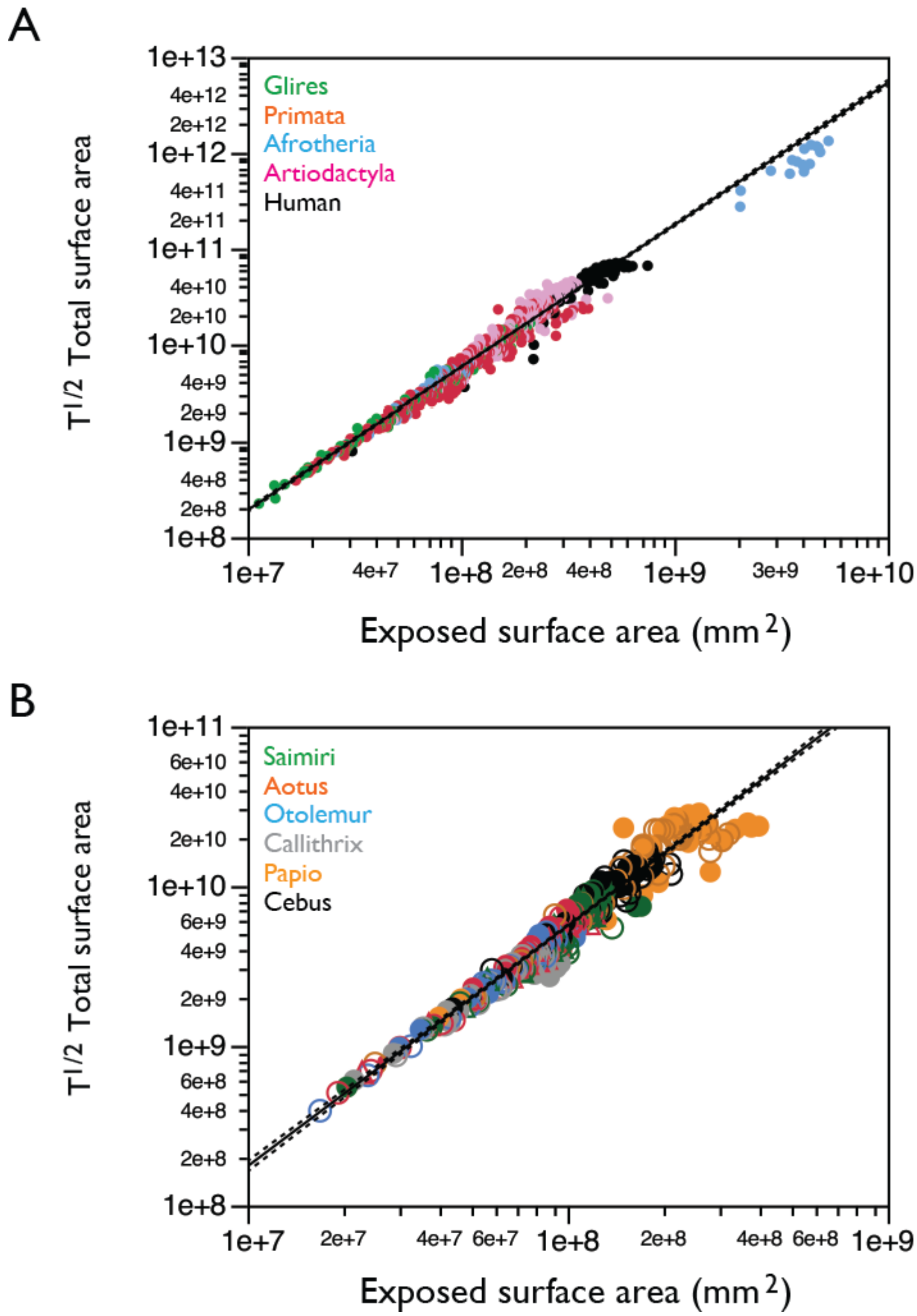
This putative (truncated) fractal shape allows us to apply a so-called Flory approximation to the surface energy term, which relies on assuming scale-invariance between the intrinsic and extrinsic sizes of a thin membrane. Importantly, at least under this mean field approximation, it is possible to show that other possible surface terms are sub-dominant to the self-avoiding term. Minimizing the total energy, one then obtains

$$\sqrt{T} A_T = k\, A_E^{5/4}$$

Crucially, the exponent 5/4 implies that the quantity k is adimensional, and thus the relation between A_T and A_E is invariant when areas are measured in units of T^2. The fundamental length scale is then $d_{min} = T/k^2$, a simple multiple of average thickness T. The transition between gyrified and non-gyrified cortices occurs simply when $A_T = A_E = d_{min}^2$; that is, when the cortical size becomes comparable to the fundamental scale, as measured in units of average thickness.

Incidentally, notice that $k \propto p + p_{cs}$, and p_{cs} is within a few percent of atmospheric pressure for all mammals in our dataset except the deep-diving cetaceans, for which p_{cs} varies with depth, but must on average be much higher than in terrestrial species. As can be seen above, a higher value of p_{cs} (and thus k) implies a greater degree of cortical folding. This might explain the apparently still slightly excessive folding in cetacean cortices compared to other mammals shown in Figure 3a.

Fig. S1.



The product $T^{1/2}$ A_G scales as a similar function of A_E across sites of the cerebral cortex within individuals and across individuals of the same species. a, each data point corresponds to one coronal section of a complete antero-posterior series of sections across the cerebral cortex of one individual of the following species: human (black), non-human primates (red), glires (green), afrotheria (blue, including the elephant) and artiodatyls (pink). The function plotted for all data points, which include smooth cortical sections, has an exponent of 1.482±0.008 (r^2=0.975, p<0.0001). b, primate data shown in a but with individuals of a same species labeled in different markers of the same color. The function plotted for all data points has an exponent of 1.506±0.016 (r^2=0.956, p<0.0001). T, A_G and A_E measured in mm and mm^2. Data from (*15-17, 19-21*).

Table S1.

Datasets used in this study. All values refer to one cortical hemisphere only.

	Our dataset			Other datasets			
Species	A_G, mm^2	**FI**	T, mm	A_G, mm^2	**FI**	T, mm	Reference
Marsupialia							
Marmota mitis				219	**1.000**	0.806	1
Didelphis virginiana				660	**1.119**	0.826	1
Wallabia rufogrisea				2,165	**1.230**	2.471	1
Macropus melanops				3,745	**1.413**	2.243	1
Afrotheria							
Procavia capensis	2,113	**1.355**	1.482				17
Dendrohyrax dorsalis	1,895	**1.360**	1.598				17
Loxodonta africana	257,067	**4.137**	2.633	313,750	**3.840**	2.234	15, 1
Setifer setosus				216	**1.000**	0.944	1
Elephantulus fuscipes				233	**1.000**	0.865	1
Potamogale velox				414	**1.000**	1.618	1
Rhychocyon stuhlmanni				715	**n.a.**	1.406	1
Trichechus manatus				6,300	**1.060**	4.230	23
Glires							
Mus musculus	148	**1.019**	0.725				20
Rattus norvegicus	356	**1.043**	1.212	322	**1.000**	n.a.	20, 1
Cavia porcellus	536	**1.044**	1.515				20
Dasyprocta promnolopha	1,413	**1.171**	1.846				20
Hydrochoerus hydrochoeris	6,005	**1.304**	2.814				20
Oryctolagus cuniculus				690	**1.150**	n.a.	1
Dolychotus sp.				3,255	**1.418**	2.012	1
Scandentia							
Tupaia glis				645	n.a.	0.946	1
Tupaia belangeri	497	**1.041**	1.070				19
Primata							
Callimico goeldii	1,953	**1.262**	1.600				19
Callithrix sp.				2,020	**1.224**	1.507	1
Callithrix jacchus	1,534	**1.175**	1.327				19
Otolemur garnettii	1,745	**1.247**	1.428				19
Aotus trivirgatus	2,214	**1.356**	1.488	1,890	**n.a.**	1.971	19, 1
Saimiri sciureus	5,250	**1.574**	1.468	3,980	**1.598**	1.671	19, 1
Cebus apella	7,653	**1.694**	1.761				19
Galagoides demidoff				675	**n.a.**	0.946	1
Perodicticus potto				1,990	**n.a.**	1.927	1

Cercopithecus aethiops				10,000	**1.852**	1.800	1
Macaca mulatta				12,500	**1.701**	2.204	1
Macaca radiata	8,441	**1.807**	1.566				19
Macaca fascicularis	9,381	**1.652**	1.411				19
Papio anubis cynocephalus	16,689	**1.919**	2.032				19
Homo sapiens	94,050	**2.210**	2.938	121,500	**2.826**	2.811	10, 21
Homo sapiens				113,750	**2.862**	n.a.	1
Eulipotyphla							
Sorex minutus				40	**1.000**	0.367	1
Sorex araneus				70	**1.000**	0.388	1
Crocidura russala				61	**1.000**	0.438	1
Crocidura flavescens				97	**1.000**	0.619	1
Erinaceus europaeus				462	**1.000**	1.047	1
Neomys fodiens				106	**1.000**	0.415	1
Talpa europaea				194	**1.000**	0.858	1
Galemys pyrenaicus				247	**1.000**	0.996	1
Carnivora							
Felis catus				4,165	**1.593**	1.837	1
Procyon lotor				6,100	**1.848**	n.a.	1
Vulpes vulpes				6,750	**1.985**	n.a.	22
Vulpes vulpes				7,550	**1.804**	1.815	1
Canis latrans				11,350	**1.802**	n.a.	1
Artiodactyla							
Sus scrofa domesticus	6,594	**1.889**	1.336	18,000	**2.169**	2.189	16, 1
Antidorcas marsupialis	9,623	**2.018**	1.669				16
Damaliscus dorcas	16,915	**2.008**	2.169				16
Tragelaphus stripceros	22,203	**2.026**	2.344				16
Giraffa camelopardalis	40,128	**2.408**	2.394				16
Bos taurus				52,750	**2.488**	2.142	1
Ovis aries				14,900	**1.935**	1.738	1
Perissodactyla							
Equus caballus				57,550	**n.a.**	1.990	24
Equus caballus				61,400	**2.797**	2.296	1
Cetacea							
Phocaena phocaena				65,000	**4.127**	1.577	1
Stenella styx				73,700	**n.a.**	3.087	1
Delphinus delphis				85,850	**3.993**	n.a.	1
Tursiops truncatus				135,000	**4.762**	1.715	1
Grampus griseus				156,600	**4.290**	1.992	1
Globicephala				290,750	**5.239**	2.024	1

macrorhyncha							
Pseudorca crassidiens				739,200	**9.935**	n.a.	1

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